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Waymo is hitting the highway – but can it handle the speed? / Life is a highway, and Waymo needs to learn to drive it if it wants to grow.

by [+ Andrew J. Hawkins](#)

Nov 12, 2025, 1:00 PM GMT-3



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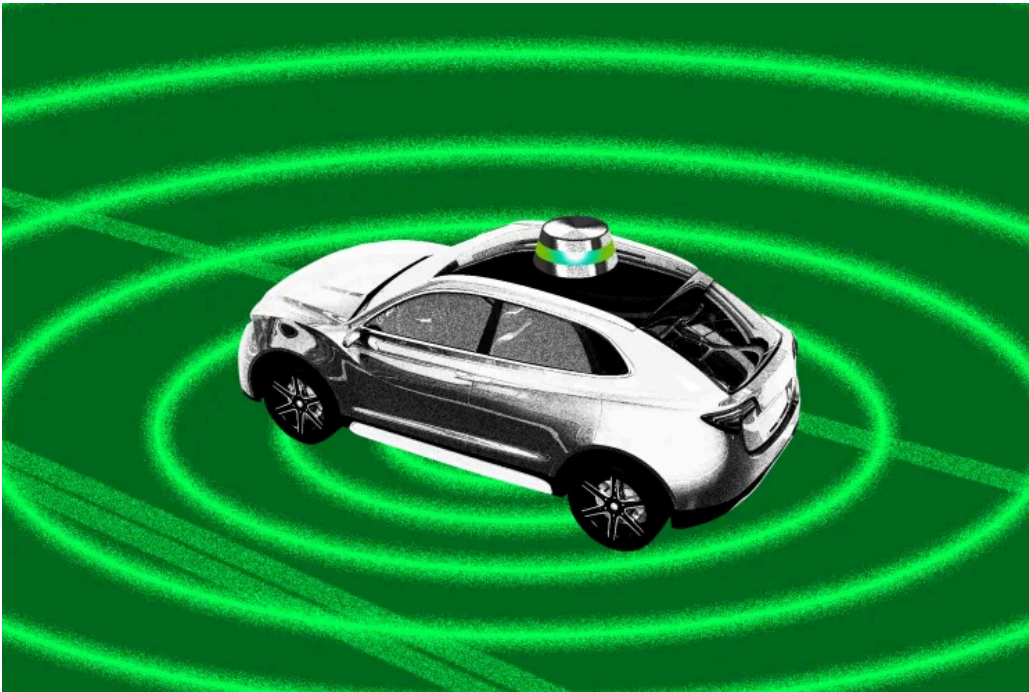
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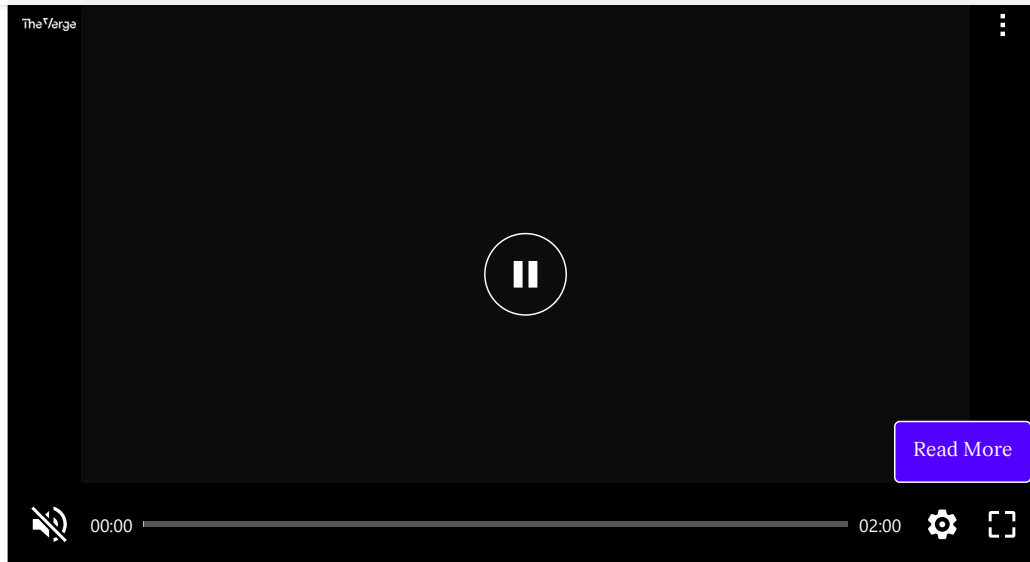
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Waymo is finally ready to hit the highway. Starting today, the company's robotaxis will gradually start to include more highway trips in its routes in Phoenix, San Francisco, and Los Angeles. In addition, Waymo's Bay Area service is extending south to San Jose, including 24/7 curbside access at both terminals of San Jose International Airport — the company's second airport service after Phoenix.

has not gone unnoticed by customers, who often note their trips can take longer because the vehicles are prohibited from using routes that travel on highways. But after years of testing, including on public highways with employees as well as on closed courses and in virtual simulation, Waymo says it's ready to start offering highway trips to a lot more people.

"Freeway driving is one of those things that's very easy to learn, but very hard to master when we're talking about full autonomy without a human driver as a backup," Dmitri Dolgov, co-CEO of Waymo, said in a briefing with reporters. "And at scale. So it took time to do it properly with a strong focus on system safety and reliability."





Waymo will start conducting highway trips in Phoenix, Los Angeles, and the San Francisco Bay Area. Initially, only early-access users — those who have opted in to test new Waymo features — will be able to take trips that include freeway travel. Over time, the feature will expand gradually to more riders as performance data and feedback are collected. From the rider’s perspective, the experience will stay familiar. Users hail a ride via the Waymo app, view their ETA and route preview, and if the freeway route is significantly faster, the system automatically selects it.

Waymo is hoping riders will reward the service with higher ratings, as freeway routing can make trips up to 50 percent faster, such as from San Francisco to Mountain View, while also helping connect riders more efficiently to public transit, and improving “first-mile” and “last-mile” mobility.

The challenges of highway driving are numerous. Higher traffic speeds mean Waymo’s autonomous vehicles will have less time to make consequential decisions. Any mistake can carry a higher degree of severity. The company’s engineers say

“Freeway driving is one of those things that’s very easy to learn, but very hard to master when we’re talking about full autonomy without a human driver as a backup.”

Because freeways can be so much more challenging, Waymo has built more redundancy into its systems to account for a wider variety of edge cases, including simulating a total power failure to one of Waymo’s dual onboard computers. In that scenario, the system immediately activates its backup, allowing the vehicle to maintain control and safely navigate to the nearest freeway exit. Pierre Kreitmann, principal software engineer at the company, likens this to a human suddenly losing half their vision and brainpower but still driving safely.

When Waymo vehicles do need to pull over, the company says it has well-established protocols to keep riders safe and ensure their trips can continue. Waymo is coordinating closely with the Arizona Department of Public Safety, California Highway Patrol, and other regional safety authorities to ensure readiness and alignment with local rules and regulations.

Highways have been a top target for Waymo for many years now. Critics have cited the avoidance of highways as evidence that autonomous vehicles aren’t ready for the realities of driving long distances. Self-driving truck companies have largely stuck to highways in their testing, though usually with human safety drivers in the front seat.

Waymo is unique insofar as it is one of the only robotaxi companies offering trips in fully driverless cars. Tesla operates a “robotaxi” service in California with safety drivers behind the wheel that’s only open to select riders, which apparently includes highway routes.

to San Francisco International Airport (SFO), which makes San Jose's Mineta Airport the company's first official airport in California.

There will be dedicated curbside pickup and drop-off zones at both Terminals A and B. Meanwhile, SFO is still in the early pilot stage. Waymo received its permit to start commercial operations at SFO in September, but it's still coordinating with airport officials to phase in operations gradually. The company has been locked in negotiations with SFO for several years as it seeks to assure regulators that its vehicles can handle their chaotic environments, with thousands of cars, taxis, shuttles, and passengers constantly intermixing every day.

Airports and highways are inexorably linked, as most airports are accessed through highway driving. They're also huge cash cows for ridehail companies, accounting for an estimated 20 percent of human-driven services like Uber and Lyft. Waymo will need to master both airports and highways if it wants to successfully compete with traditional ridehailing — to say nothing of making a profit.

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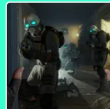
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